

Swiss Restaurant Report

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1 Introduction

The following report is based on a data set provided by the Swiss Link market research institute, which contains scores given to different questions about the experience of clients in a restaurant. The institute surveyed 1033 subjects, and data set covers 18 questions (the questionnaire was composed of more questions, but the data were pre-processed by a third-party in order to make the data summarized). The answers are given on a scale from 1 (least possible score) to 10 (highest possible score). Moreover, options "11" and "12" were available to the respondent, respectively meaning "does not apply" and "no answer given".

The 18 questions are defined by a numerical code (06, 07, 08, 09, 13, 14, 15, 16) whose attribution is determined by the main subject of the question (for example service quality, food quality, location quality etc...). Each sub-question thoroughly analyzes a particular feature of the restaurant. An additional question, g27a, represents the general satisfaction with the restaurant, and could be seen as a response variable whose outcome can be explained by the other variables.

Our primary goal is to determine which are the main influences of the variable of interest, g27a. Additionally, if possible, we would like to evaluate the correlation of the variables and find out if the model could be expressed by fewer variables (i.e., if the model can be explained by factors that group some variables reducing its dimensions).

2 Data Visualization

By looking at the data, the first striking feature is the high quantity of missing values, defined by responses "11" and "12". Out of 1033 responses, only 442 subjects have responded to all the questions on the scale of 1 to 10. I deemed the number of complete cases too small with respect to the total number of responses, so I analyzed the missing values with the ultimate idea of imputing missing data.

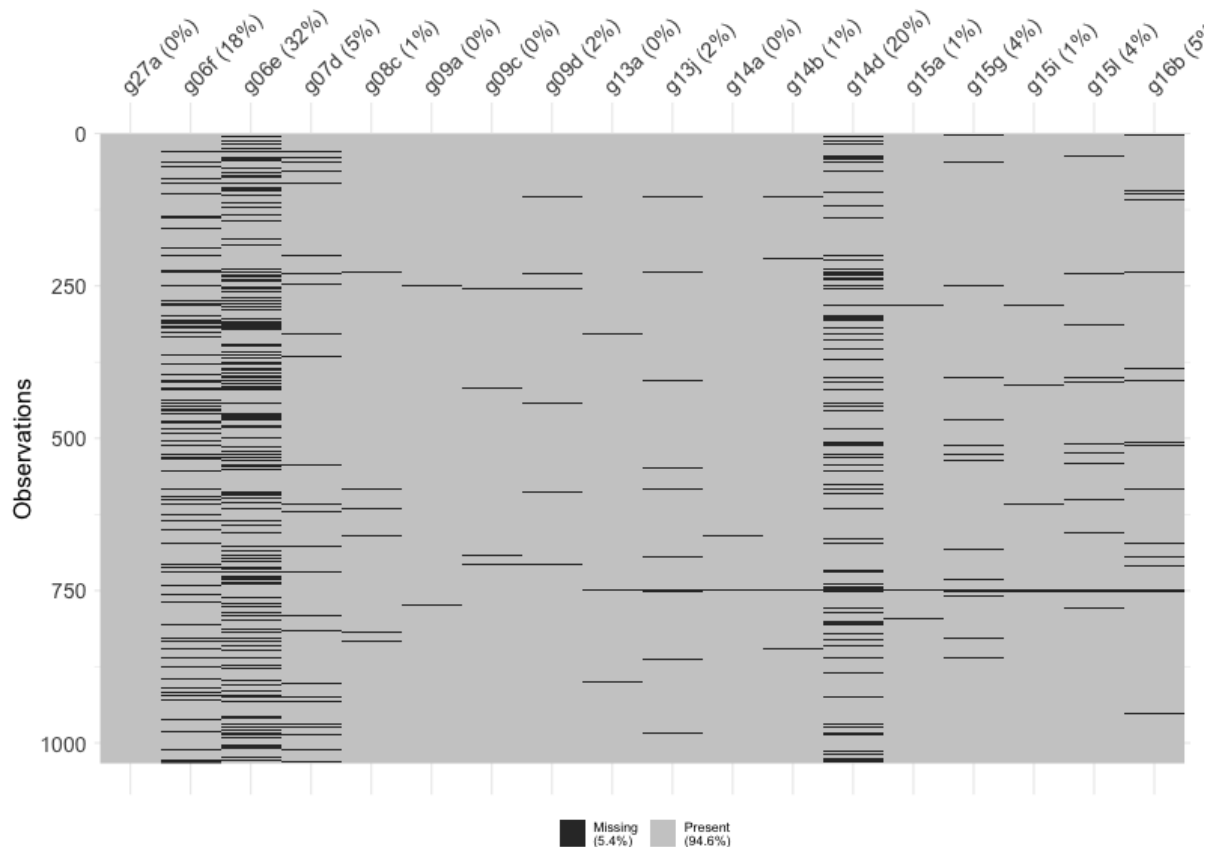


Figure 1: Percentage of missing values per variable.

In the graph above, made with the "naniar" R package, we can see that some variables have a lot more missing values than others (g06f, g06e and g14d). If the data were missing completely randomly, it would be strange to see so many missing values in only 3 categories and we would expect to see them spread out more evenly throughout all the variables.

In order to have a bit of a clearer idea on how the missing data are, I have performed a test using TestMCARNormality in R, which tests whether the missing data could be of the MCAR class, but the null hypothesis was rejected with a very small p-value (0.00001) which indicates that missing values are not MCAR.

Furthermore, using a paired t-test, I tested whether the values of g27a are significantly different from the values of g27a where g06f is missing, and the result further indicates that the missing values were not randomly missing.

Given those considerations, I decided to perform data imputing in order to replace missing values with plausible ones. Specifically, I employed the MICE package available on R, and I carried out imputation using "pmm" (Predictive Mean Matching). The data set to which I will refer is made up of observed values and PMM imputed values from the first imputed data set.

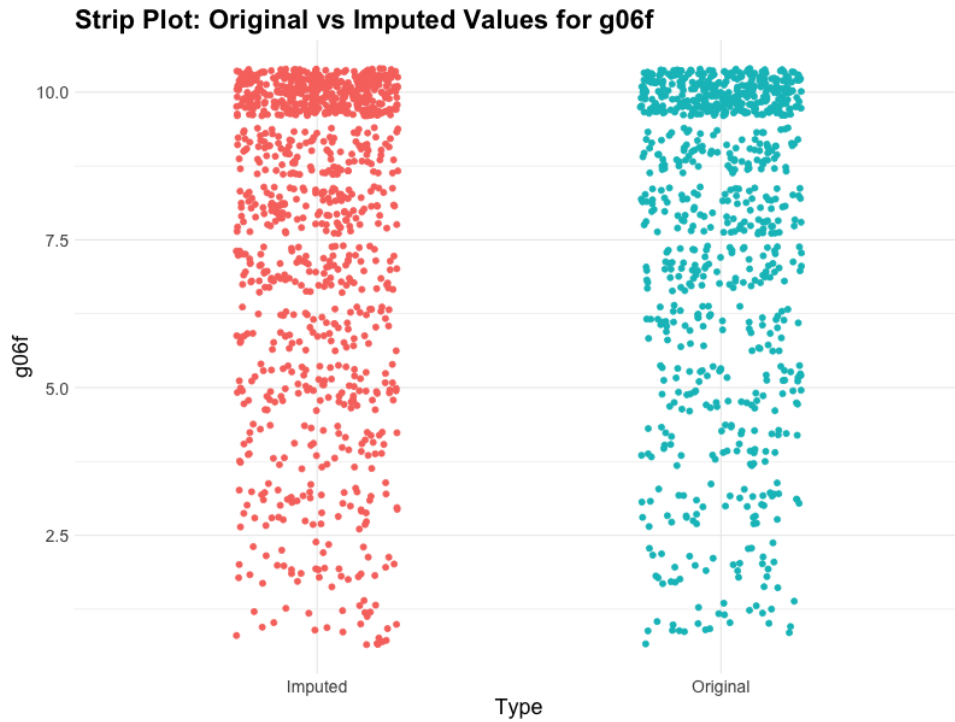


Figure 2: Strip Plot comparing imputed and original values for g06f (using jitter).

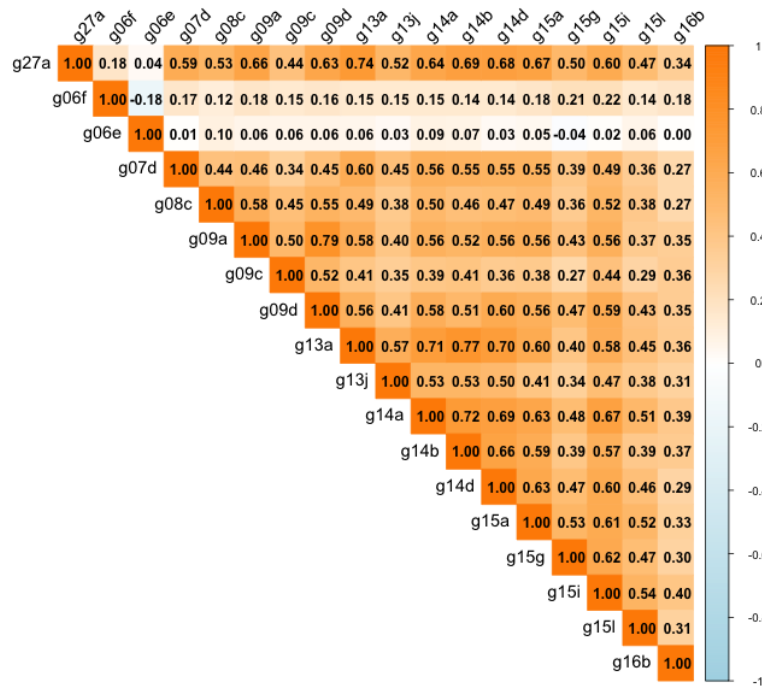


Figure 3: Correlation matrix of the variables, after having imputed the data.

By looking at the correlation matrix, it is possible to notice that we have very different values. For all questions, an increase in the score leads to an increase in general satisfaction, but some variables (g06f, g06e) seem to have very little effect, probably due to the specificity of the questions (parking and public transport situation).

The same variables also have nearly zero correlation with the other questions, particularly g06e. We have some high correlation values, especially with questions that belong to the same group (for example, $\text{Corr}(g09a, g09d)=0.79$), but there are no strikingly high numbers that indicate almost certain multicollinearity. In general, we have moderate correlation between all the variables.

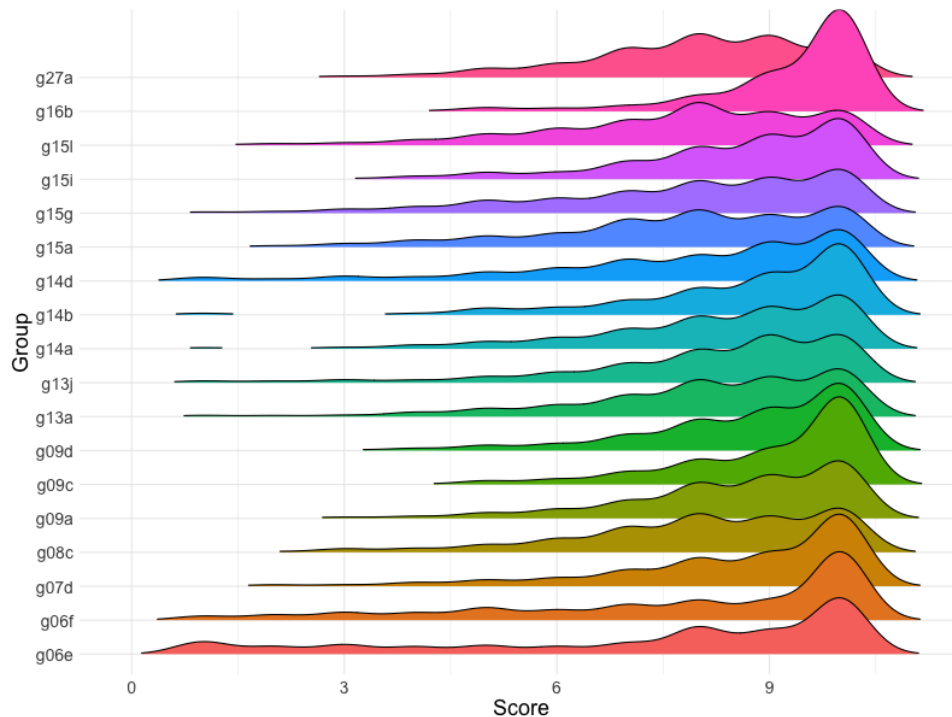


Figure 4: Ridgeplot graph of all variables, after having imputed the data.

The ridgeplot above tells us that the response variable g27a seems pretty spread out between 6 and 10, with peaks around 9 but that do not stand out from the distribution. It does also seem that very few people gave very low scores, with the majority of respondents being sufficiently happy with their experience (from 6/10 onwards). Question g16b (correctness of bill) is the one where the restaurant scored the highest proportion of outstanding scores, with a high peak at 10, followed by g09c (meals large enough).

Generally speaking, all questions have most of their density after 7, showcasing that the restaurant does not seem to have a weak point. The group of questions where the restaurant scored the lowest seems to be group 15, while the question that had the biggest difference of judgment between the respondents is g06e, obtaining a lot of very high scores but also a significant amount of low scores.

Since the questions are labeled with respect to the main subject they are trying to evaluate (e.g., group 09 assesses the general quality of the food), we can suggest that it is better to combine the variables and to evaluate g27a using the 8 different groups (instead of 17 variables). Therefore, we can try to visualize the changes in the correlation matrix and in the ridgeplot if we consider the groups as the variables (model-wise, I will analyze it later on). Group values have been calculated as the arithmetic mean of the values of the singular questions making up each group, for every observation.

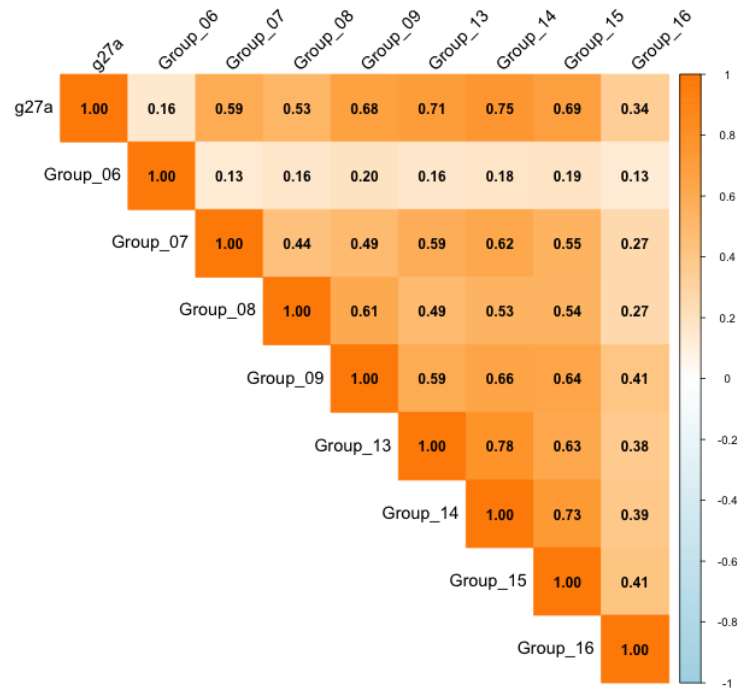


Figure 5: Correlation matrix of grouped variables, after having imputed the data.

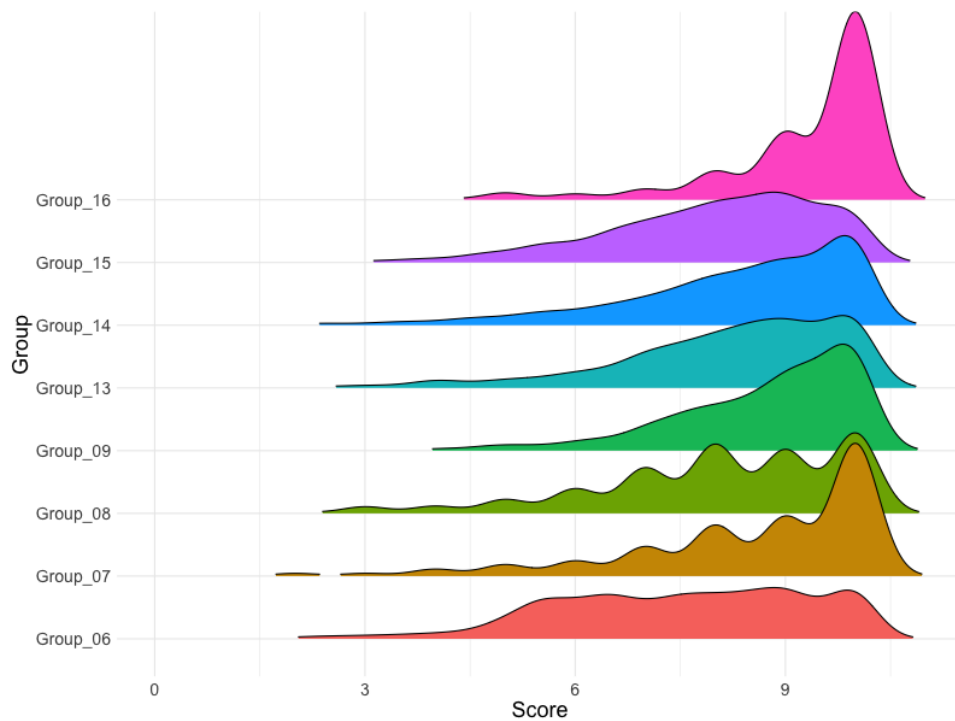


Figure 6: Ridgeline plot of grouped variables, after having imputed the data.

By looking at the correlation matrix, we can see that Group 06 is still very low-correlated with the other variables, but it is not close to zero like it was for g06e in the full model. As far as the other correlations are concerned, there are no differences to what we could expect by looking at the full-model correlation matrix, since we obtained pretty similar numbers overall. The ridgeline plot shows that some of the groups are more evenly distributed with respect to the single-question distributions separately, with almost no peaks especially in Group 6 and in Group 15. Group 7, 8 and 16 are still the same since they are made of one question only, and the rest of the groups still have very high density near 9 indicating that, of the groups made by more than one question, they were generally rated the highest overall.

3 Statistical Analyses

I started the analysis by fitting an OLS linear model with separate variables, in order to evaluate both the model itself and the presence of leverages and outliers. After using the "lm" function, we can see that a lot of variables are not significant (for example, g06e, g06f, g07d, g15i). Furthermore, after computing Cook's Distances for the model observations, it is clear that we have a lot of influential units, some of them being pretty striking.

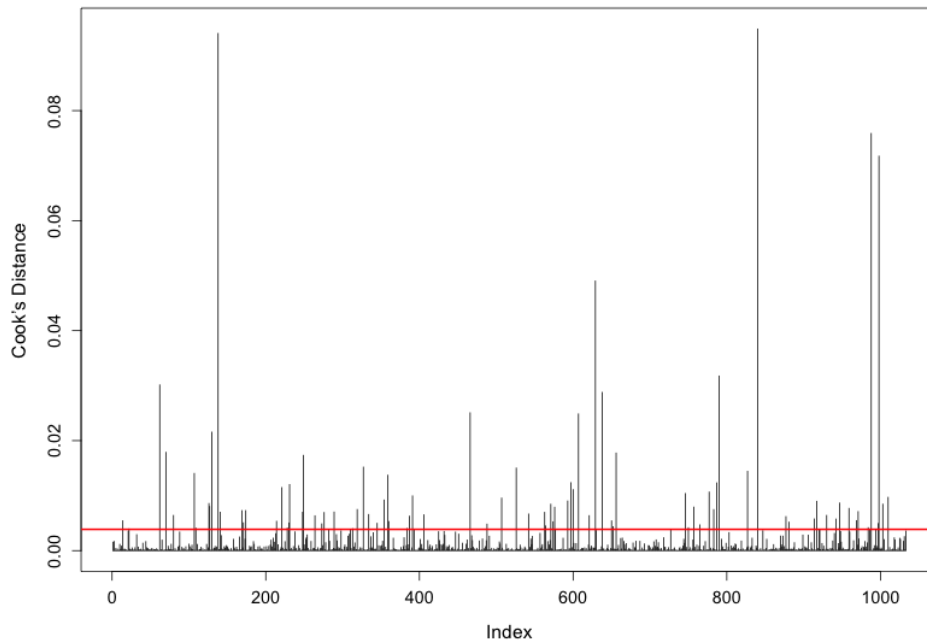


Figure 7: Cook's distances of the OLS linear model including all the variables.

Therefore, I have decided to perform robust regression on both the separate data set and the grouped data set.

In particular, denoting as 1 the full RR model and as 2 the grouped RR model, we can notice that:

- Model 1 has the highest Multiple and Adjusted R-squared (0.7634, 0.7595), with Model 2 having 0.7380 and 0.7359, respectively. Therefore, according to R-squared, the full RR model is the best fit.

-Model 1 has the lowest residual standard error (0.7342), with Model 2 having 0.7714. Therefore, the residual standard error indicates that Model 1 has a higher predictive accuracy. Additionally, robust regression highlighted some outliers and the majority of the observations have weights different from 1, though not far off that value.

Given these reasons, I decided that the best way to explain the data set was through the full-variable robust model. Analyzing it, we can see that the variables that were statistically significant at 5% were:

1. g07d (coefficient: 0.082684, p-value: 0.00119);
2. g09a (coefficient: 0.170346, p-value: 8.97e-07);
3. g13a (coefficient: 0.213970, p-value: 1.40e-08);
4. g14b (coefficient: 0.147395, p-value: 4.10e-05);
5. g14d (coefficient: 0.067547, p-value: 0.00683);
6. g15a (coefficient: 0.143989, p-value: 2.22e-07);
7. g15g (coefficient: 0.050131, p-value: 0.00973);
8. g15l (coefficient: 0.049713, p-value: 0.03527).

No question from Group 6 was found to be statistically significant, nor was g16b. Question g08c was significant at 10% level (coefficient: 0.036891, p-value: 0.08673). After evaluating the VIFs, no significant sign of multicollinearity was found in the model.

Subsequently, I shifted the analysis on the reduction of the dimensions of the data set. Therefore, I have fitted two factor models (the first using 2 factors, the second using 3 factors) and compared them to see which one best represented the data set. The differences between the 2 factor models are not very evident, but almost all of them seem to point to the fact that the 2-factor model would be preferable in this case.

The cumulative proportion of variance explained in the 2-factor model is 48%, whereas in the 3-factor model is 51%. Given that 48% is definitely not outstandingly high, but can be generally considered quite good, a 3% increase does not compensate for the high increase of complexity obtained in the 3-factor model.

The RMSR value for both models is satisfying, being less than 0.05 and therefore indicating good fits.

In the 3-factor model, PA3 (third factor) seems to have very low explaining power, and looks very weak compared to PA1 and PA2.

In conclusion, the 2-factor model was deemed the most suitable factor model with respect to this data set.

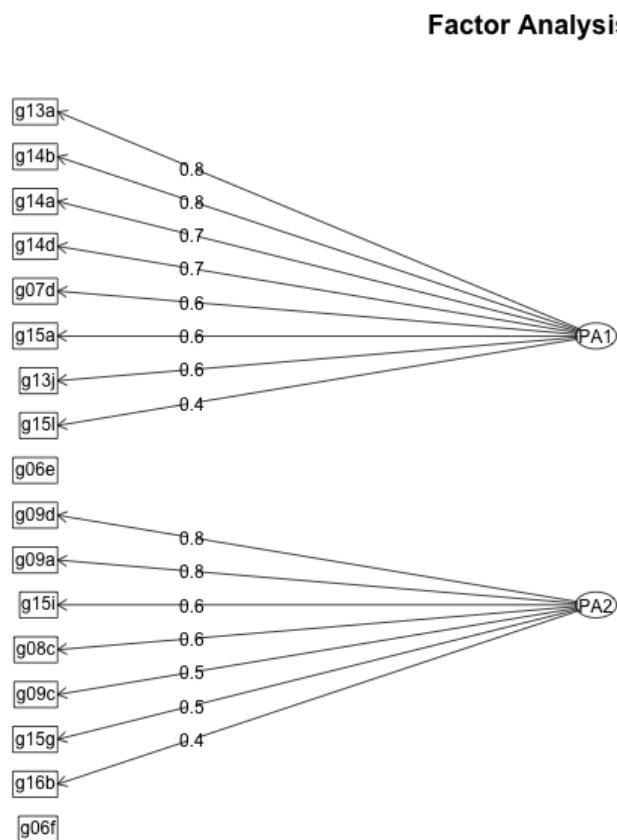


Figure 8: Factor model diagram, indicating stronger loading for each variable.

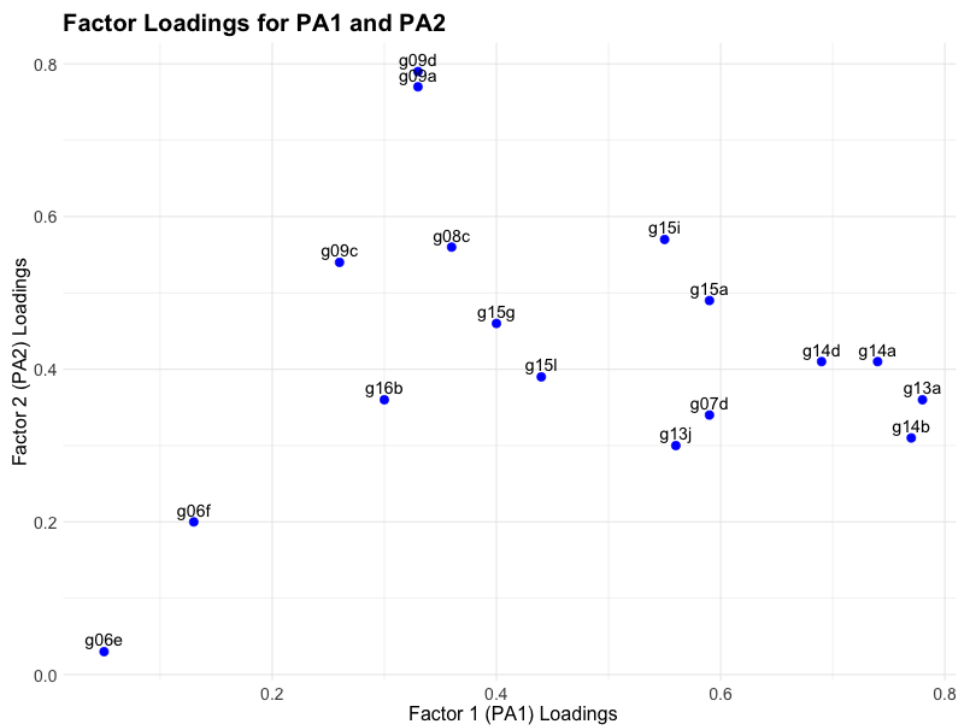


Figure 9: Factor model biplot, showcasing each variable with respect to both loadings.

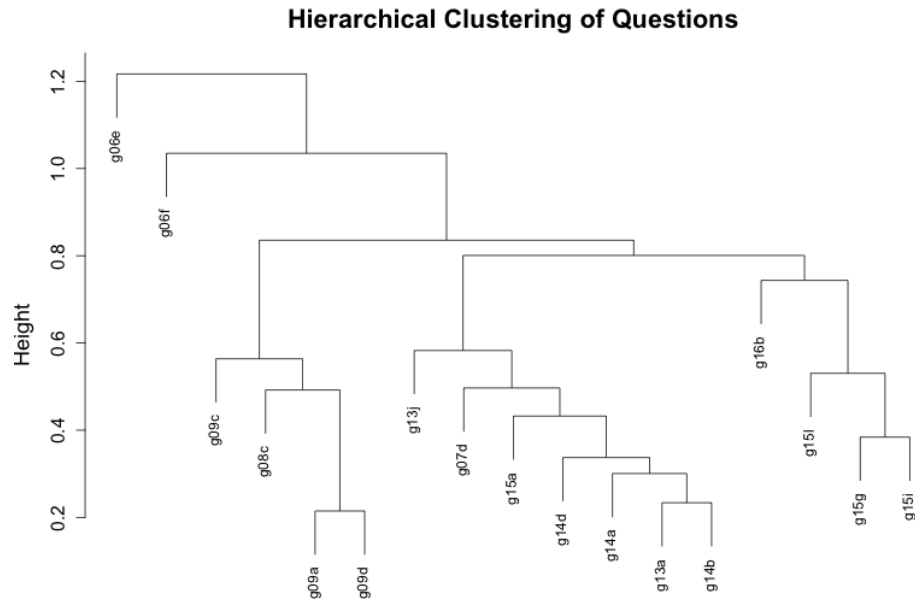


Figure 10: Hierarchical clustering based on the distance matrix of the correlations between the questions.

Using factor model loadings, plots and hierarchical clustering, it is possible to try and group the questions beyond their natural group of belonging. The first plot indicates which is the highest loading for each variable; the second plot shows how every question behaves with respect to both loadings at the same time; the third plot is a dendrogram that displays how variables can be clustered using empirical observations, evaluating the distance matrix of correlations between the questions. After summarizing the three plots above, I have reached my conclusions:

1. Questions g07d, g13a, g13j, g14a, g14b, g14d and g15a can be grouped together. They refer to the quality of the service that can be imputed to people working at the restaurant (e.g., kindly placed at good table, general service, waiting time, friendliness, etc.).
2. Questions g15i, g15g, g15l and g16b can be grouped together. They generally refer to the quality of the restaurant with respect to "objects", not to people (e.g., chairs, cleanliness, air quality, bill).
3. Questions g08c, g09a, g09c and g09d can be grouped together. They refer to the quality of the dinner with respect to what can be eaten, not the restaurant or the service per se (e.g., menu choice, taste and quantity of food, quality of ingredients).

Questions belonging to Group 06 have shown very different behaviour, not being included in any cluster and having very little to almost no effect on the outcome of the response variable g27a.

4 Particular Features of the Data

After having fitted the full robust model, we obtained that 4 observations were considered outliers: n°138,607,629 and 840. In all four cases, the general satisfaction score was much lower than what we could expect given the questions scores. Since for all such cases we have a majority of very high scores and some particularly low ones, we can make the case that the experience with respect to the low score features was so disappointing that it visibly dragged down the general evaluation. This is a personal interpretation, and thus should be just taken as an hypothesis since we cannot investigate further. The model has attributed a weight equal to zero to all four of the observations.

5 Conclusion for the Client

As a result of this data analysis, I recommend to the restaurant to focus primarily on the features that most significantly affect the general satisfaction, and those are: taste of food, quality of general service, friendliness and atmosphere in the restaurant. Variables like parking spots, public services, meals large enough and cleanliness have been found as the least significant ones.

Moreover, it is possible to conclude that the questions can be grouped in three main clusters (human service quality, non-human service quality, food quality), considerably reducing the dimensions of the data set.

6 Appendix

```
library(tidyverse)
library(performance)
library(faraway)
library(MASS)
library(corrplot)
library(corr)
library(rsample)
library(dplyr)
library(car)
library(ggplot2)
library(tidyr)
restdata <- read.table(file.choose(),header=TRUE)
restdata[restdata == 11 | restdata == 12] <- NA

#g27a General satisfaction with restaurant
#g06f Good parking situation
#g06e Good to access by public transport
#g07d Kindly placed at good table
#g08c Choice on the menu
#g09a Taste of food
#g09c Meals large enough
#g09d Quality of ingredients
#g13a Quality of general service
#g13j Waiting time before payment
```

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#g14b Friendliness
#g14a Neat appearance of waiters
#g14d Competent information about food and drinks
#g15a Atmosphere in the restaurant
#g15g Air quality/freshness
#g15i Restaurant clean
#g15l Comfortable chairs
#g16b Correctness of bill

#####LM#####
modell1 <- lm(g27a ~ ., data = restdata)
summary(modell1)
cormatrix <- cor(restdata, use = "pairwise.complete.obs")
library(corrplot)
corrplot(
  cormatrix,
  method = "color",
  type = "upper",
  addCoef.col = "black",
  number.cex = 0.8,
  tl.col = "black",
  tl.srt = 45,
  tl.cex = 1,
  col = colorRampPalette(c("lightblue", "white", "darkorange"))(200),
  mar = c(0, 0, 2, 0)
)

vif_values <- vif(modell1)
print(vif_values)
check_model(modell1)
check_outliers(modell1)
library(mice)
md.pattern(restdata[,2:18])

#Number of rows with no missing data
complete_rows <- restdata %>%
  filter(complete.cases(.)) %>%
  nrow()

#What kind of missingness?
library(naniar)
png("missingvalues.png", width=800, height=600)
p <- vis_miss(restdata)
p + theme(axis.text.x = element_text(size = 14),
          axis.text.y = element_text(size = 14),
          axis.title = element_text(size = 16),
          plot.margin = margin(9, 9, 9, 9))
dev.off()
tapply(restdata$g27a, is.na(restdata$g06f), mean, na.rm = TRUE)
t.test(restdata$g27a ~ is.na(restdata$g06f))
library(MissMech)

```

```

TestMCARNormality(restdata)
library(mice)

for(i in 1:ncol(restdata)) {
  restdata[, i][is.na(restdata[, i])] <- mean(restdata[, i], na.rm = TRUE)
}
print(restdata)
model2 <- lm(g27a ~ ., data = restdata) #mean imputation
summary(model2)
imputed_data1 <- mice(restdata, method = "norm.predict", m = 1)
print(imputed_data1$data)
imputed_data2 <- mice(restdata, method = "pmm", m = 1, seed=500) #chosen method
impdata <- complete(imputed_data2, 1)
strip_data <- data.frame(
  Value = c(restdata$g06f, impdata$g06f),
  Type = rep(c("Original", "Imputed"), each=length(restdata$g06f))
)

png("stripplot.png",width=800,height=600)
ggplot(strip_data, aes(x=Type, y=Value)) +
  geom_jitter(aes(color=Type), width=0.2, size=2) +
  labs(title="Strip Plot: Original vs Imputed Values for g06f", y="g06f") +
  theme_minimal() +
  theme(plot.title = element_text(size = 22, face = "bold"),
        axis.title.x = element_text(size = 18),
        axis.title.y = element_text(size = 18),
        axis.text.x = element_text(size = 14),
        axis.text.y = element_text(size = 14),
        legend.position="none")
dev.off()

model3 <- lm(g27a ~ ., data = impdata)
summary(model3)
vif_values <- vif(model3)
print(vif_values)
check_model(model3)
check_outliers(model3)

cormatrix <- cor(impdata)
library(corrplot)
png("corplotswiss.png",width=800, height=600)
corrplot(
  cormatrix,
  method = "color",
  type = "upper",
  addCoef.col = "black",
  number.cex = 1,
  tl.col = "black",
  tl.srt = 45,
  tl.cex = 1.2,
  col = colorRampPalette(c("lightblue", "white", "darkorange"))(200),

```

```

    mar = c(0, 0, 2, 0)
  )
dev.off()

implong_data <- pivot_longer(impdata, cols=everything(),
                             names_to = "Group", values_to = "Score")

library(ggribes)
png("ridge.png", width=800, height=600)
ggplot(implong_data, aes(x = Score, y = Group, fill = Group)) +
  geom_density_ridges(scale = 3, rel_min_height = 0.01) +
  labs(x = "Score", y = "Group") +
  theme_minimal() +
  theme(
    axis.title.x = element_text(size = 18),
    axis.title.y = element_text(size = 18),
    axis.text.x = element_text(size = 14),
    axis.text.y = element_text(size = 14),
    legend.position = "none")
dev.off()

#Residuals vs Fitted Values
plot(model3$fitted.values, resid(model3),
     main="Residuals vs Fitted", xlab="Fitted Values", ylab="Residuals")
abline(h=0, col="red", lwd=2)

#Q-Q Plot
qqnorm(resid(model3), col="red", lwd=2)

#Histogram of Residuals
hist(resid(model3), breaks=30, main="Histogram of Residuals", col="lightblue")

#Scale-Location Plot
plot(model3$fitted.values, sqrt(abs(resid(model3))),
     main="Scale-Location Plot", xlab="Fitted Values", ylab="sqrt|Residuals|")
abline(h=0, col="red", lwd=2)

#Cooks Distance Plot
png("cooks.png",width=800, height=600)
plot(cooks.distance(model3), type="h", ylab="Cooks Distance",cex.main=1.5,
     cex.lab=1.3, cex.axis=1.2)
abline(h=4/(length(resid(model3))), col="red", lwd=2)
dev.off()

library(robustbase)
roblm<-lmrob(g27a~., data=impdata, estimator="MM")
summary(roblm)
#Residuals vs Fitted Values
png("check1.png",width=800,height=600)
plot(roblm$fitted.values, resid(rlmrobust_model),

```

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    main="Residuals vs Fitted Values", xlab="Fitted Values",
      ylab="Residuals",cex.main=1.5,cex.lab=1.3, cex.axis=1.2)
abline(h=0, col="red", lwd=2)
dev.off()
#Q-Q Plot
png("check2.png",width=800,height=600)
qqnorm(resid(roblm), col="red", lwd=2,cex.main=1.5, cex.lab=1.3, cex.axis=1.2)
dev.off()
#Histogram of Residuals
png("check3.png",width=800,height=600)
hist(resid(roblm), breaks=30, main="Histogram of Residuals", col="lightblue",
      cex.main=1.5,cex.lab=1.3, cex.axis=1.2)
dev.off()

#Scale-Location Plot
png("check4.png",width=800,height=600)
plot(roblm, which = 3, main = "Scale-Location Plot", cex.main=1.5,cex.lab=1.3,
      cex.axis=1.2)
dev.off()

#####FA#####
library(psych)
impdata<-imputed_data2$data
impdata_no_g27a <- impdata[, !(names(impdata) %in% "g27a")]
R_imp <- cor(impdata_no_g27a)

imp_fa1vmax <- fa(r = R_imp, nfactors = 2, fm = "pa", rotate = "varimax")
#2-factor model
print(imp_fa1vmax)
png("fadiagram.png", width=800, height=600)
fa.diagram(imp_fa1vmax,
  main = "Factor Analysis Diagram",
  cex.main = 1.5,
  cex.lab = 1.3,
  cex.axis = 1.2)
dev.off()
imp_fa2vmax <- fa(r = R_imp, nfactors = 3, fm = "pa", rotate = "varimax")
#3-factor model
print(imp_fa2vmax)
fa.diagram(imp_fa2vmax)

imp_fa3vmax <- fa(r = R_imp, nfactors = 4, fm = "pa", rotate = "varimax")
#4-factor model
print(imp_fa3vmax)
fa.diagram(imp_fa3vmax)

library(ggplot2)
library(ggrepel)
library(plotly)

```

```

loadings_df <- data.frame(
  Variable = c("g06f", "g06e", "g07d", "g08c", "g09a", "g09c", "g09d", "g13a",
               "g13j", "g14a", "g14b", "g14d", "g15a", "g15g", "g15i", "g15l",
               "g16b"),
  PA1 = c(0.13, 0.05, 0.59, 0.36, 0.33, 0.26, 0.33, 0.78, 0.56, 0.74, 0.77,
          0.69,
          0.59, 0.40, 0.55, 0.44, 0.30),
  PA2 = c(0.20, 0.03, 0.34, 0.56, 0.77, 0.54, 0.79, 0.36, 0.30, 0.41, 0.31,
          0.41,
          0.49, 0.46, 0.57, 0.39, 0.36))

print(loadings_df)
png("fact.png",width=800,height=600)
ggplot(loadings_df, aes(x = PA1, y = PA2, label = Variable)) +
  geom_point(color = "blue", size = 3) +
  geom_text(vjust = -0.5, hjust = 0.5, size = 5) +
  labs(title = "Factor Loadings for PA1 and PA2",
       x = "Factor 1 (PA1) Loadings",
       y = "Factor 2 (PA2) Loadings") +
  theme_minimal() +
  theme(
    plot.title = element_text(size = 20, face = "bold"),
    axis.title.x = element_text(size = 16),
    axis.title.y = element_text(size = 16),
    axis.text.x = element_text(size = 14),
    axis.text.y = element_text(size = 14)
  )
dev.off()

#####CLUSTER#####

dist_questions <- as.dist(1 - cor(impdata_no_g27a))

#Hierarchical clustering for questions
hc_questions <- hclust(dist_questions, method = "ward.D2")
png("clusttt.png",width=800,height=600)
plot(hc_questions, main = "Hierarchical Clustering of Questions", xlab = "",
     sub = "",cex.main = 1.8,
     cex.lab = 1.5,
     cex.axis = 1.3)
dev.off()
library(factoextra)

#####USING GROUPED VARIABLES#####

#Calculate the mean of all the questions forming a Group, and create new
#variable Group_.. with the mean
Group_06 <- rowMeans(impdata[c("g06f", "g06e")], na.rm = TRUE)

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Group_07 <- rowMeans(impdata[c("g07d")], na.rm = TRUE)
Group_08 <- rowMeans(impdata[c("g08c")], na.rm = TRUE)
Group_09 <- rowMeans(impdata[c("g09a", "g09c", "g09d")], na.rm = TRUE)
Group_13 <- rowMeans(impdata[c("g13a", "g13j")], na.rm = TRUE)
Group_14 <- rowMeans(impdata[c("g14a", "g14b", "g14d")], na.rm = TRUE)
Group_15 <- rowMeans(impdata[c("g15a", "g15g", "g15i", "g15l")], na.rm = TRUE)
Group_16 <- rowMeans(impdata[c("g16b")], na.rm = TRUE)
grouped_means <- data.frame(
  Group_06 = Group_06,
  Group_07 = Group_07,
  Group_08 = Group_08,
  Group_09 = Group_09,
  Group_13 = Group_13,
  Group_14 = Group_14,
  Group_15 = Group_15,
  Group_16 = Group_16
)
grouped_means <- cbind(g27a = impdata$g27a, grouped_means)
head(grouped_means)

groupR<-cor(grouped_means)
png("groupcorr.png",width=800, height=600)
corrplot(
  groupR,
  method = "color",
  type = "upper",
  addCoef.col = "black",
  number.cex = 1,
  tl.col = "black",
  tl.srt = 45,
  tl.cex = 1.2,
  col = colorRampPalette(c("lightblue", "white", "darkorange"))(200),
  mar = c(0, 0, 2, 0)
)
dev.off()

grouped_means_no_g27a <- grouped_means[, !(names(grouped_means) %in% "g27a")]
Rgroupfa<-cor(grouped_means_no_g27a)
groupfa1vmax <- fa(r = Rgroupfa, nfactors = 2, fm = "pa", rotate = "varimax")
#2-factor model
print(groupfa1vmax)
fa.diagram(groupfa1vmax)

groupfa2vmax <- fa(r = Rgroupfa, nfactors = 3, fm = "pa", rotate = "varimax")
#3-factor model
print(groupfa2vmax)
fa.diagram(groupfa2vmax)

library(robustbase)
groupobl<-lmrob(g27a~., data=grouped_means)
summary(groupobl)

```



```

#Residuals Plot
residuals <- residuals(grouproblm)
plot(residuals, main = "Residuals of the Robust Regression Model", ylab =
  "Residuals", xlab = "Index")
abline(h = 0, col = "red")

#Leverage and Influence
leverage <- hatvalues(grouproblm)
plot(leverage, main = "Leverage of Observations", ylab = "Leverage", xlab =
  "Index")

#Q-Q plot for residuals
qqnorm(residuals)
qqline(residuals, col = "red")

#Histogram of residuals
hist(residuals, main = "Histogram of Residuals", xlab = "Residuals", col =
  "lightblue", breaks = 30)

#Check for Multicollinearity (VIF)
library(car)
vif(grouproblm)

#Plot Fitted vs Actual
fitted_values <- fitted(grouproblm)
plot(fitted_values, grouped_means$g27a, main = "Fitted vs Actual", ylab =
  "Actual g27a", xlab = "Fitted values")
abline(0, 1, col = "red")

corre_matrix <- cor(grouped_means_no_g27a)

grouplong_data <- pivot_longer(grouped_means, cols=everything(),
  names_to = "Group", values_to = "Score")
grouplong_data_filtered <- grouplong_data %>%
  filter(Group != "g27a")

png("ridgegroup.png", width=800, height=600)
ggplot(grouplong_data_filtered, aes(x = Score, y = Group, fill = Group)) +
  geom_density_ridges(scale = 3, rel_min_height = 0.01) +
  labs(x = "Score", y = "Group") +
  theme_minimal() +
  theme(
    plot.title = element_text(size = 22, face = "bold"),
    axis.title.x = element_text(size = 18),
    axis.title.y = element_text(size = 18),
    axis.text.x = element_text(size = 14),
    axis.text.y = element_text(size = 14),
    legend.position = "none")
dev.off()

```
